



Bahir Dar University

Bahir Dar Institute of Technology

Faculty of Computing

DEPARTEMENT OF SOFTWARE ENGINEERING

Proposal submission

Name	ID
1. Bezawit Mamo	BDU1405561
2. Betelhem Tsega	BDU1404679
3. Marta Ejigu	BDU1404214

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Bahir Dar University, Bahir Dar Institute of Technology

Title 1: AI-Powered Internship Progress Verification & Performance Analytics System for Bahir Dar University

Introduction:

Internships are a core component of higher education, giving students real-world experience and essential skills. However, Bahir Dar University currently lacks a reliable way to monitor student progress during internships. Advisors and faculty cannot see what students truly accomplish each week, whether company-provided grades are fair, if students are developing the expected skills, or if mentors provide adequate guidance. This often results in inconsistent evaluations, missed learning opportunities, and uncertain skill acquisition. This project proposes an AI-powered Internship Progress Monitoring and Analytics System that tracks student activities, verifies skill development, evaluates mentor support, and provides evidence-based reports to university advisors, ensuring transparency, fairness, and measurable growth.

Statement of the Problem

The university faces challenges in monitoring internships due to:

- **Unreliable Progress Reports:** Current methods rely on manual submissions or self-reports, which may not reflect actual work done.
- **Limited Oversight:** Advisors cannot see whether students are actively engaged or meeting internship objectives.
- **Delayed Feedback:** Faculty and supervisors often learn about student performance too late to provide timely guidance.
- **Low Accountability:** Students may not consistently log their activities, leading to incomplete evaluation and poor learning outcomes.

These issues reduce the effectiveness of internships and hinder skill development and university-industry collaboration.

Objectives

The goal of this project is to implement a system that verifies and visualizes student internship progress in real-time:

- **Student Activity Logging:** Develop a secure platform where students record tasks, projects, or time spent on internship activities.
- **AI Progress Verification:** Use AI algorithms to analyze activity data, detect inconsistencies, and verify that students are working as intended.
- **Supervisor & Advisor Dashboard:** Provide university faculty and internship supervisors with real-time dashboards showing verified student engagement, task completion, and skill development.

- Automated Reports & Alerts: Generate evidence-based progress reports and notifications for underperforming students, enabling timely intervention

Significance of the Project

- Ensures transparent and reliable monitoring of student internship work.
- Provides university advisors and supervisors with real-time, evidence-based updates.
- Improves accountability and motivation among students.
- Enables data-driven decision-making for internship evaluation and grading.
- Establishes a scalable model for monitoring internships across other faculties or universities in Ethiopia

Title 2: PassionAI: An AI-Powered Personalized Learning & Skill Explorer for Kids & Youth

Introduction

Many children and youth in Ethiopia lack opportunities to explore their true interests and develop skills aligned with their passions. Traditional education often follows a rigid curriculum that does not cater to individual preferences, creativity, or emerging talents. As a result, many students feel unmotivated, miss opportunities to discover their potential, and struggle to identify career-relevant skills early. PassionAI proposes an AI-powered platform that helps kids and youth discover their passions, explore skill-based learning paths, and access personalized guidance through interactive content, AI-generated exercises, and mentorship. The system provides a structured yet flexible learning environment that fosters creativity, self-directed growth, and practical skill development.

Statement of the Problem

Children and youth face challenges in discovering and nurturing their talents due to:

- Lack of personalized learning guidance that aligns with individual interests.
- Limited access to mentors or skill-based resources, especially outside urban centers.
- One-size-fits-all curricula that fail to motivate students or highlight practical skills.
- Difficulty in tracking skill progression or receiving timely feedback.

These gaps reduce engagement, limit skill development, and delay career exploration opportunities for young learners.

Objectives

The goal of PassionAI is to provide an AI-powered platform that empowers youth to learn according to their passions while tracking skill growth and connecting with mentors:

- **Personalized Learning Pathways:** Implement AI algorithms to analyze student preferences, skill levels, and interests to recommend customized learning activities.
- **Skill Assessment & Progress Tracking:** Provide interactive exercises, mini-projects, and AI-generated feedback to track development and highlight strengths and gaps.
- **Mentorship Integration:** Connect students with local instructors, online experts, or peer mentors based on skills and interest areas, with AI-assisted matching.
- **Resource Aggregation:** Integrate curated content from preloaded tutorials, open educational resources, and mentor-uploaded lessons to provide comprehensive learning opportunities.

Significance of the Project

- Helps children and youth discover and develop their passions early, increasing motivation and engagement in learning.
- Provides personalized skill development, guidance, and mentorship regardless of geographic location.
- Encourages creativity, self-directed learning, and early career exploration.
- Establishes a scalable, AI-driven model for passion-based education that can be expanded nationally.
- Supports social and educational empowerment by equipping youth with skills aligned to their interests and future workforce demands.

Title 3: Development of an Intelligent Resume Screening and Candidate Ranking System

Introduction

Recruitment and hiring in organizations can be slow, inconsistent, and biased due to manual resume evaluation. Human recruiters often face high workloads, making it difficult to screen large numbers of applicants efficiently. Moreover, evaluating candidates based on objective skill alignment, experience, and potential is challenging. This project proposes an AI-powered Resume Screening System that automatically analyzes, scores, and ranks candidates based on their CVs. The system streamlines the hiring process, reduces human bias, and ensures that employers identify the most qualified candidates quickly and fairly.

Statement of the Problem

Organizations face several recruitment challenges:

- High volume of applications: Reviewing hundreds or thousands of CVs manually is time-consuming.
- Inconsistent evaluation: Different recruiters may rate the same candidate differently.
- Difficulty matching skills to job requirements: Recruiters may overlook qualified candidates due to lack of systematic evaluation.
- Bias in hiring decisions: Manual screening may unintentionally favor certain candidates over others.

These issues lead to delayed hiring, suboptimal candidate selection, and potential loss of talent.

Objectives

The goal of this project is to develop an AI-driven platform that streamlines candidate selection and ensures objective evaluation:

- CV Upload & Parsing: Allow candidates to upload resumes, which the system parses for education, experience, skills, and certifications.
- Machine Learning Candidate Ranking: Implement AI/ML algorithms to score and rank candidates based on relevance to job requirements.
- Skill & Experience Matching: Automatically compare candidate qualifications against job criteria for objective evaluation.
- Dashboard & Analytics for Recruiters: Provide a user-friendly interface showing top-ranked candidates, skill gaps, and comparative scores.

Significance of the Project

- Saves time and effort for HR and recruitment teams.
- Provides objective, data-driven candidate evaluation to reduce bias.
- Helps organizations identify the most qualified candidates quickly.
- Introduces students to AI applications in industry-relevant contexts.
- Scalable to small businesses, NGOs, and large companies across Ethiopia.